

PROJECT ASSIGNMENT

AIM – To visualize the data in order to find the best location possible to set up for students undergoing the fields of Undergraduate, Postgraduate, and Ph.D.

DESCRIPTION – The Indian higher education system has expanded at a simply phenomenal rate during the 21st Century - and it shows no sign of stopping. This rapid growth in the number of individual UG, PG, and Ph.D. universities is also producing a range of research opportunities, ranging from cutting-edge Science and Engineering projects to unique programs exploring the country's own diverse history and culture. With the help of Tableau, the data is been analyzed which is without, a doubt, one of the leaders in the data viz space.

PROJECT STATEMENT - "Hi, I wanted to start a UG, PG and Ph.D. college but I want to set it up at the best locations possible. Can you help me find the best place for me to set up these three colleges?"

SOFTWARE USED – Tableau

Tableau is a data visualization tool. Therefore, its technology is there to support complex computations, data blending, and dashboarding for the purpose of creating beautiful visualizations that deliver insights that cannot easily be derived from staring at a spreadsheet.

DATA USED – states.csv

The text file is split into 3 sections UG, PG, and Ph.D. students, and each section is further classified as Male, female, and Total students in that field

UG – Male, Female, Total

PG – Male, Female, Total

Ph.D. – Male, Female, Total

CREATION –

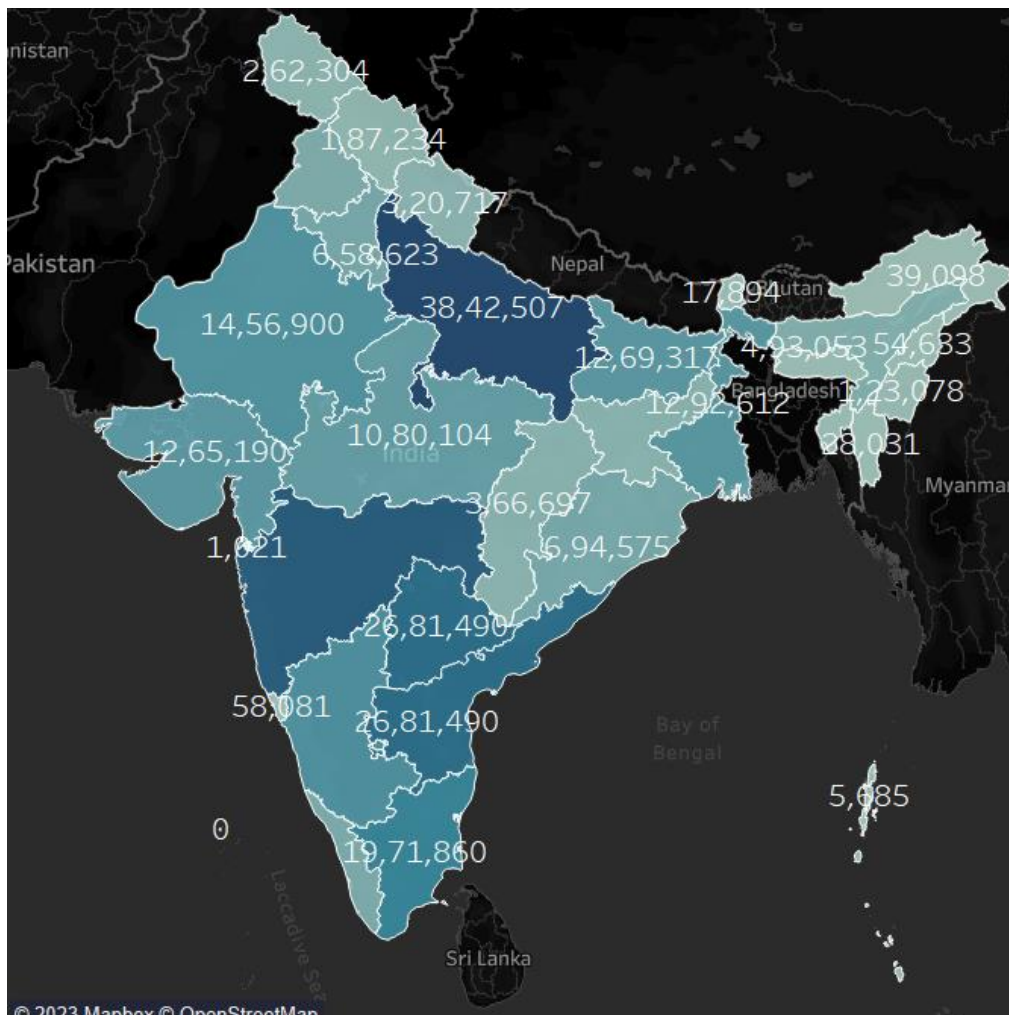
- This gives the basic idea on where is the best location to set up UG, PG, and Ph.D. colleges in India.
- Each data is analyzed individually, we need to visualize the data taken and tabulate a table in order to create a graph to picturize the data given in that particular department.
- We are given data of students in each state with their respective departments.
- Once the table for each department is created, we need to make the respective graphs because graphs can present an immense amount of data quickly and in an easy-to-consume fashion, also they are particularly useful when there is a point to be made in

the shape of the data, or for showing how different things (variables) relate to each other.

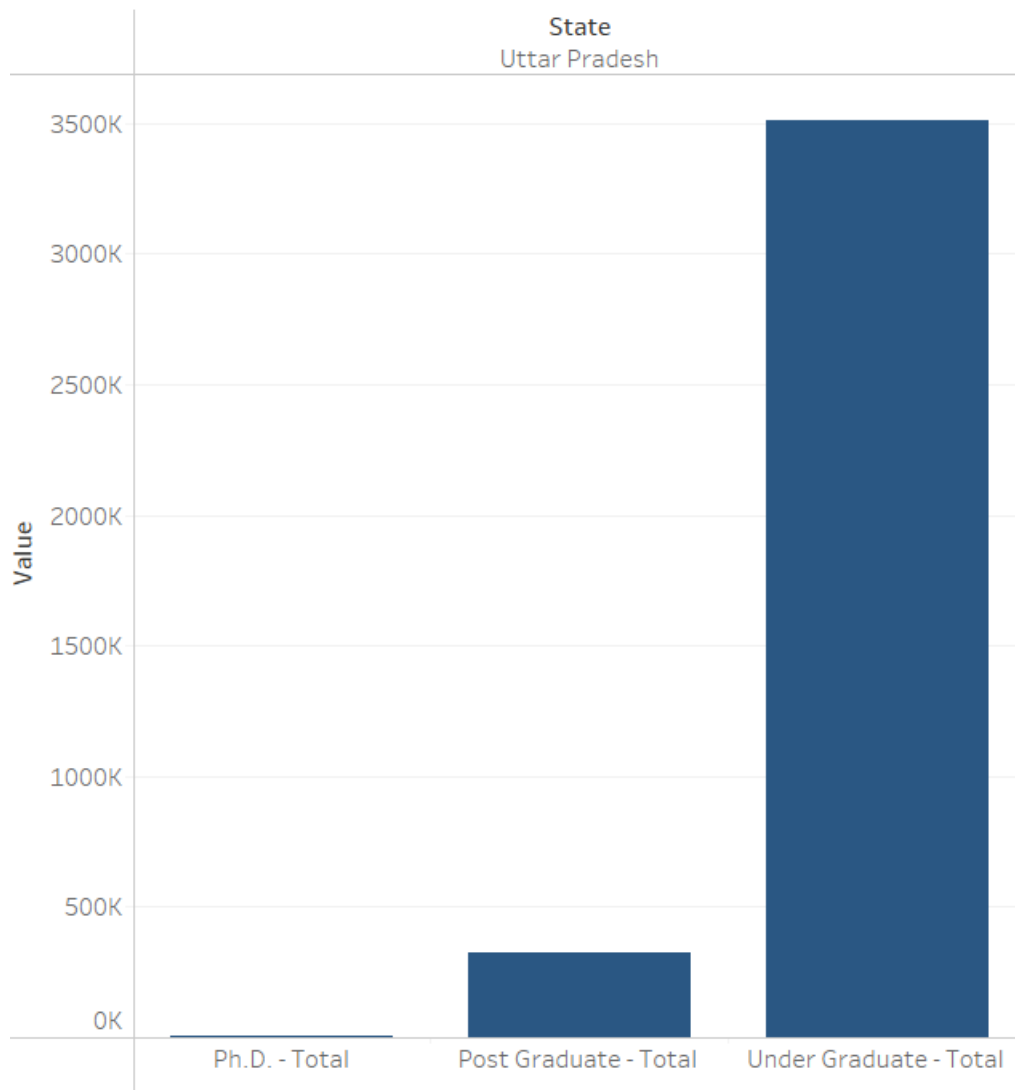
- After graphs are made, we need to use the data to pictorize in the map of India.
- Each state on the map can be colored with different colors according to the choice of the user. This will make the map look more attractive. The graph can be combined into one graph taking the subtotal of the individual divisions as one.
- Once the map is created, next is the filling of data in the map
- We need to create a new field in the graph which is the subtotal of all the students in each of the departments and fill it in the map according to each state.
- Once the data is entered, we need to set the map in the dashboard alongside the graphs created before and turn on the filter option which would show the particular data of the state when it's been clicked on the map.
- Building dashboards with Tableau allows even non-technical users to create interactive, real-time visualizations in minutes. In just a few clicks, they can combine data sources, add filters, and drill down into specific information.

DASHBOARD

In the dashboard, the left side shows the visual representation of the number of PG, UG, and Ph.D. students on the map of India. The total of each state is calculated by taking the subtotal of all the students in each field.

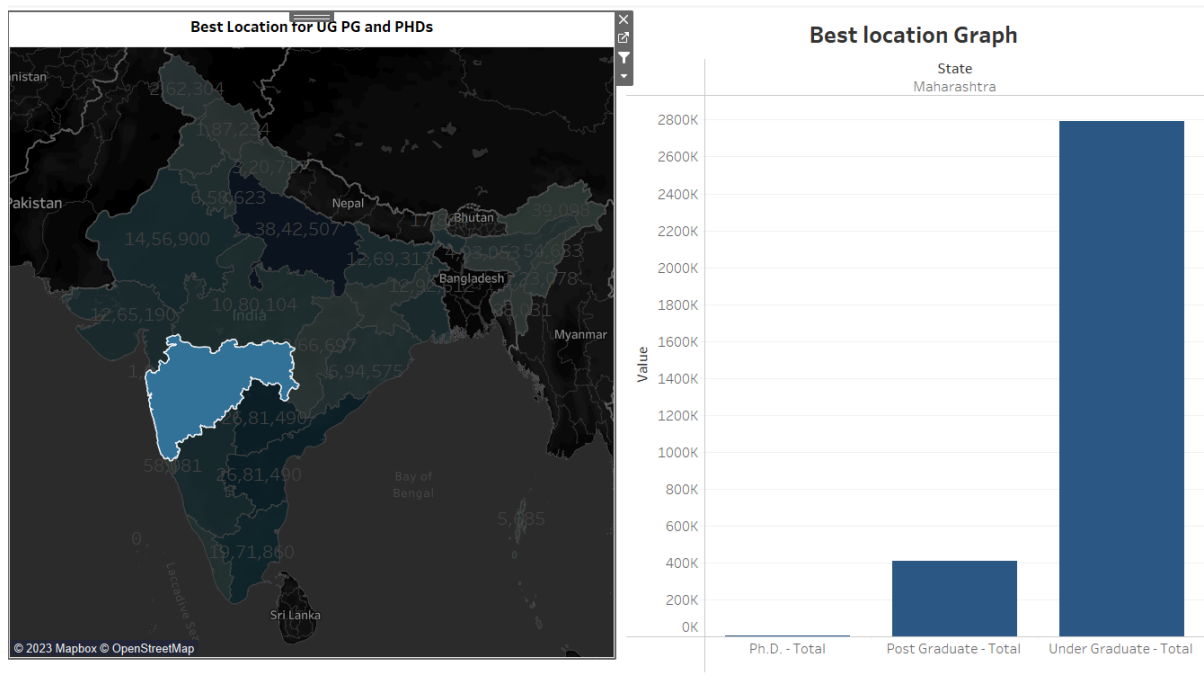


Best location Graph



- When we click on one state, all the information for the number of students is displayed.
- The graph above shows the representation of students in each field in Uttar Pradesh.
- The graph will change automatically once a different state is clicked on in the map.
- Also the values in Y axis change depending on the state clicked.

- ✚ This is how the dashboard looks when a particular state is clicked. The screenshot taken below shows how many students are undergoing each department in Maharashtra.



FINAL RESULT –

- ✚ By visualizing the data and checking all the possible combinations of variations in the graph, we discover that the best location possible to set up for students undergoing the fields of Undergraduate, Postgraduate, and Ph.D. is in Uttar Pradesh with the maximum grand total of 38,42,507 students which is the highest recorded among the other states.
- ✚ Among the top 3 locations are Uttar Pradesh, Maharashtra, and Andhra Pradesh alongside Telangana are also some contestants to set up colleges in India.

CONCLUSION – This project helped me understand how to visualize a huge set of information into simple visualizations and graphs which are more convenient to understand rather than big sets of information which would be very difficult to read and understand.